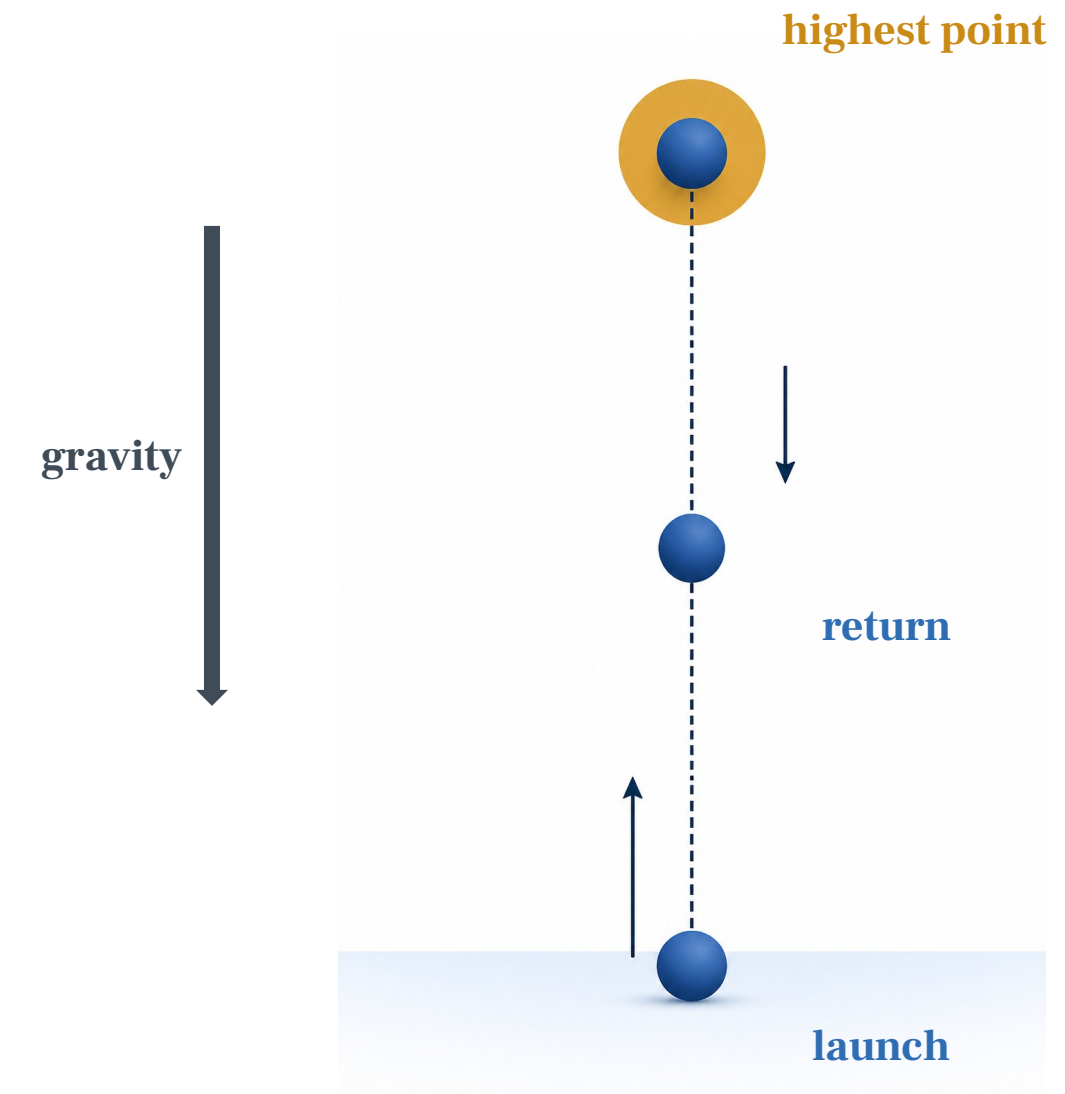


Physics to Arrays and Plots

Week 1 | From Motion to MATLAB



physical model



arrays



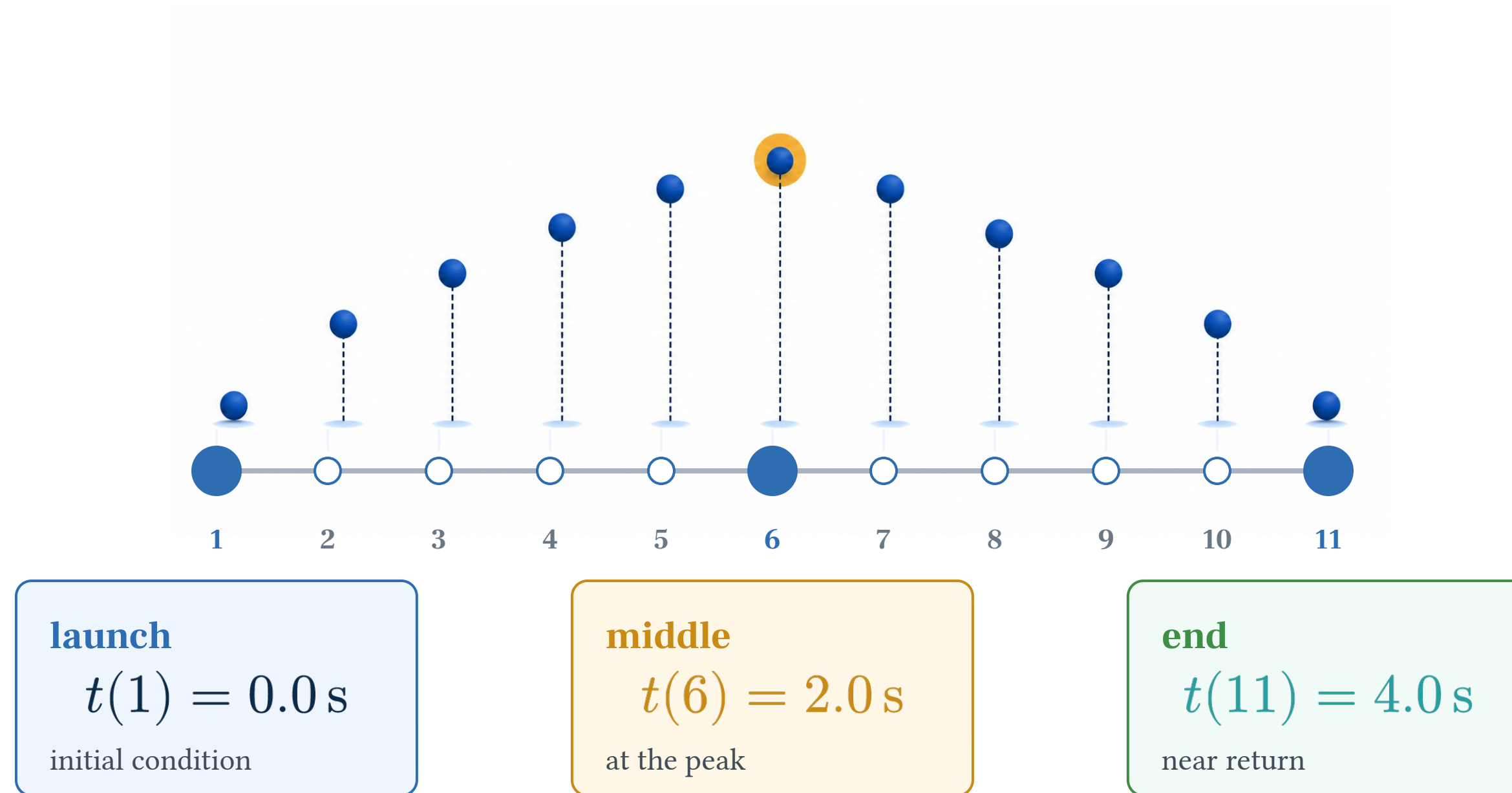
plot



validation

Arrays store physical moments

Each index identifies one measurement time



The number in parentheses selects one physical moment from the array

A labelled plot turns values into evidence

Axes, units, and shape must agree with the model

Before plotting

starts at

$$y = 0 \text{ m}$$

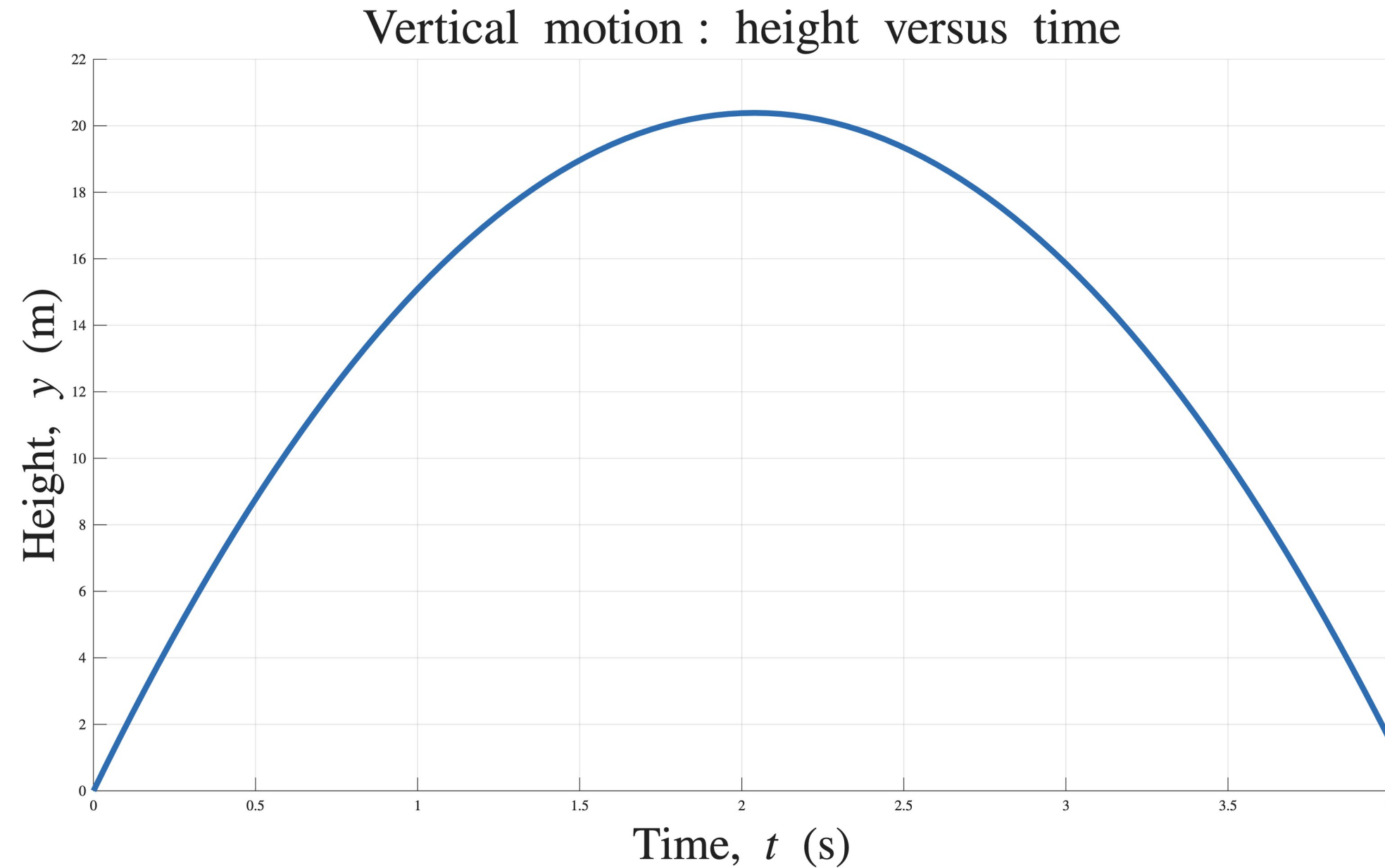
rises to

one maximum

returns to

$$y \approx 0 \text{ m}$$

prediction first

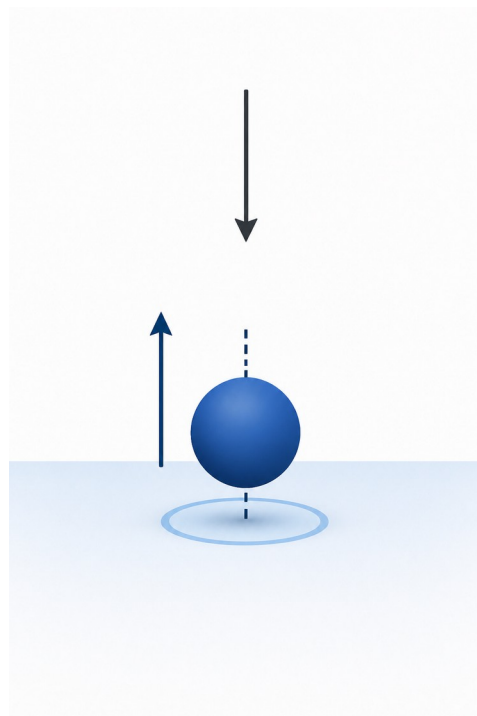


Does the curve match all three predictions?

A boundary check can reveal an error that a smooth curve hides

Use a known condition before trusting the full result

Known condition



At launch

$$t = 0 \text{ s}$$

must give

$$y = 0 \text{ m}$$

Computed result

$$y(1) = 0.000 \text{ m}$$



The first array value satisfies the launch condition

A smooth graph is not enough: test at least one value that physics already tells you

Names and units keep the model readable

Translate each physical quantity before writing code

physics symbol	meaning	MATLAB name	unit
t	time	<code>t</code>	s
y	vertical position	<code>y</code>	m
v_0	initial velocity	<code>v0</code>	m s^{-1}
g	gravitational acceleration	<code>g</code>	m s^{-2}

Readable names reduce syntax mistakes and make unit checks easier

Short code should expose the physics

The algorithm appears before the MATLAB syntax

Plain-language algorithm

- 1 Choose the model parameters
- ↓
- 2 Create an array of times
- ↓
- 3 Calculate height at every time
- ↓
- 4 Plot height against time
- ↓
- 5 Check the launch value

MATLAB

```
% model parameters
v0 = 20;    % m/s
g = 9.81;   % m/s^2

% array of times
t = linspace(0,4,11);

% height at every time
y = v0.*t - 0.5*g.*t.^2;

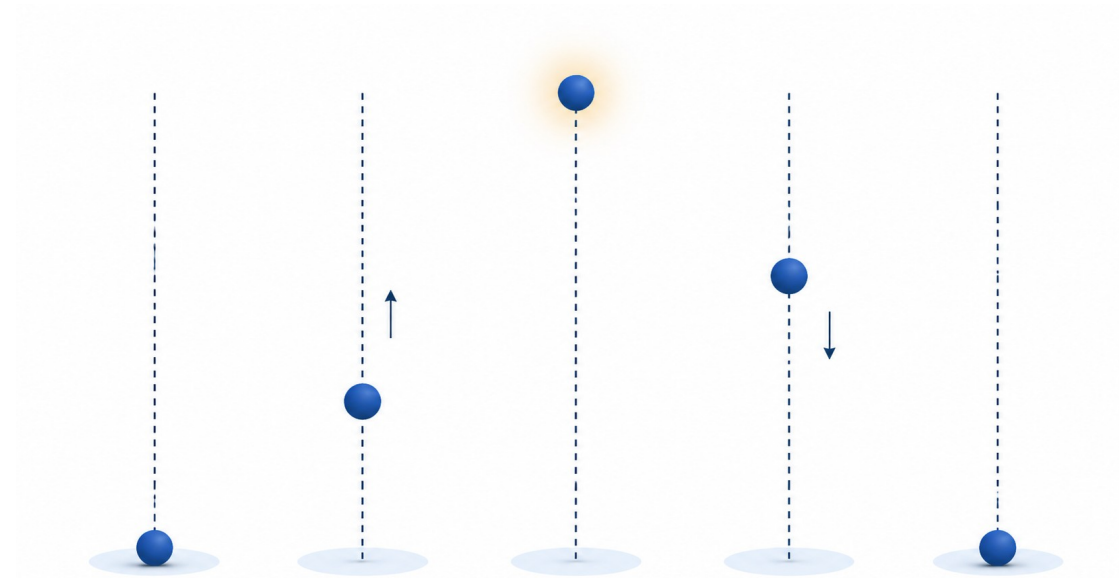
plot(t,y,'LineWidth',2)
y(1)      % boundary check
```

.* and **.^** apply the operation to every array element

The equation links each term to motion

Read the model from left to right before evaluating it

$$y(t) = y_0 + v_0t - \frac{1}{2}gt^2$$

 y_0

starting position

 v_0t

upward contribution

 $-\frac{1}{2}gt^2$

gravity contribution

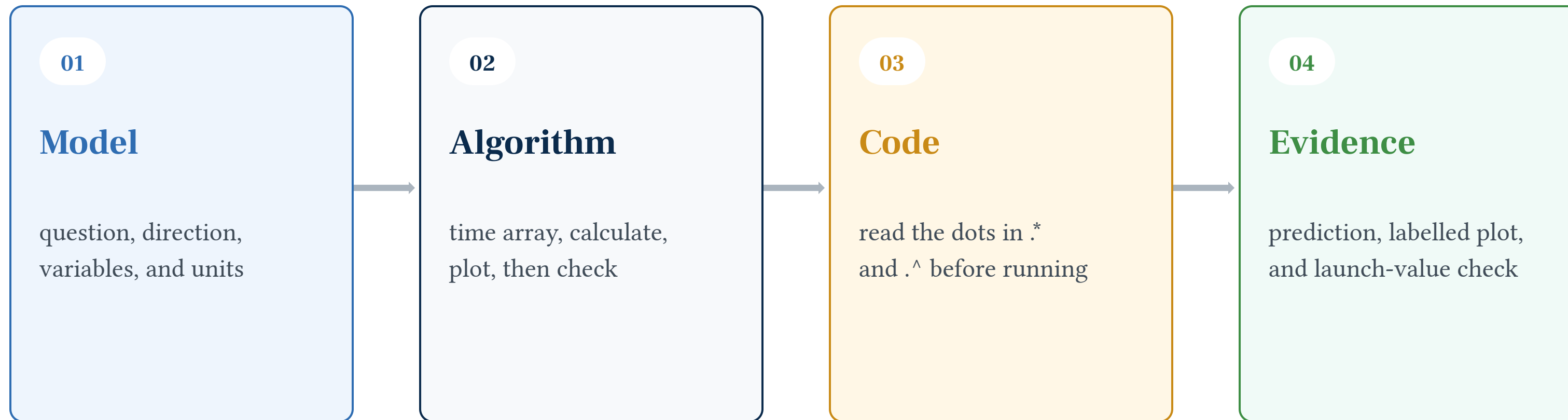
 $y(t)$

predicted position

Unit check: every term on the right must have units of metres

Trace one model from physics to a graph

Use the same reasoning chain every time you model



Checkpoint: which step would help you find a wrong sign in the model?

Build the time array first

The model needs one time value for each computed height

Two ways to make 41 samples

```
t_s = 0:0.1:4;
```

or

```
t_s = linspace(0,4,41);
```

Read one indexed value

```
length(t_s)    % 41
```

```
t_s(11)        % 1.0 s
```

MATLAB indices start at 1: t_s(1) is the launch time

Dots tell MATLAB to work through an array

Use element-wise operators when time contains many values

scalar arithmetic

`20 * 3`

one number

every time value

`v0_mps .* t_s`

one value for every time

array square

`t_s.^2`

every time value squared

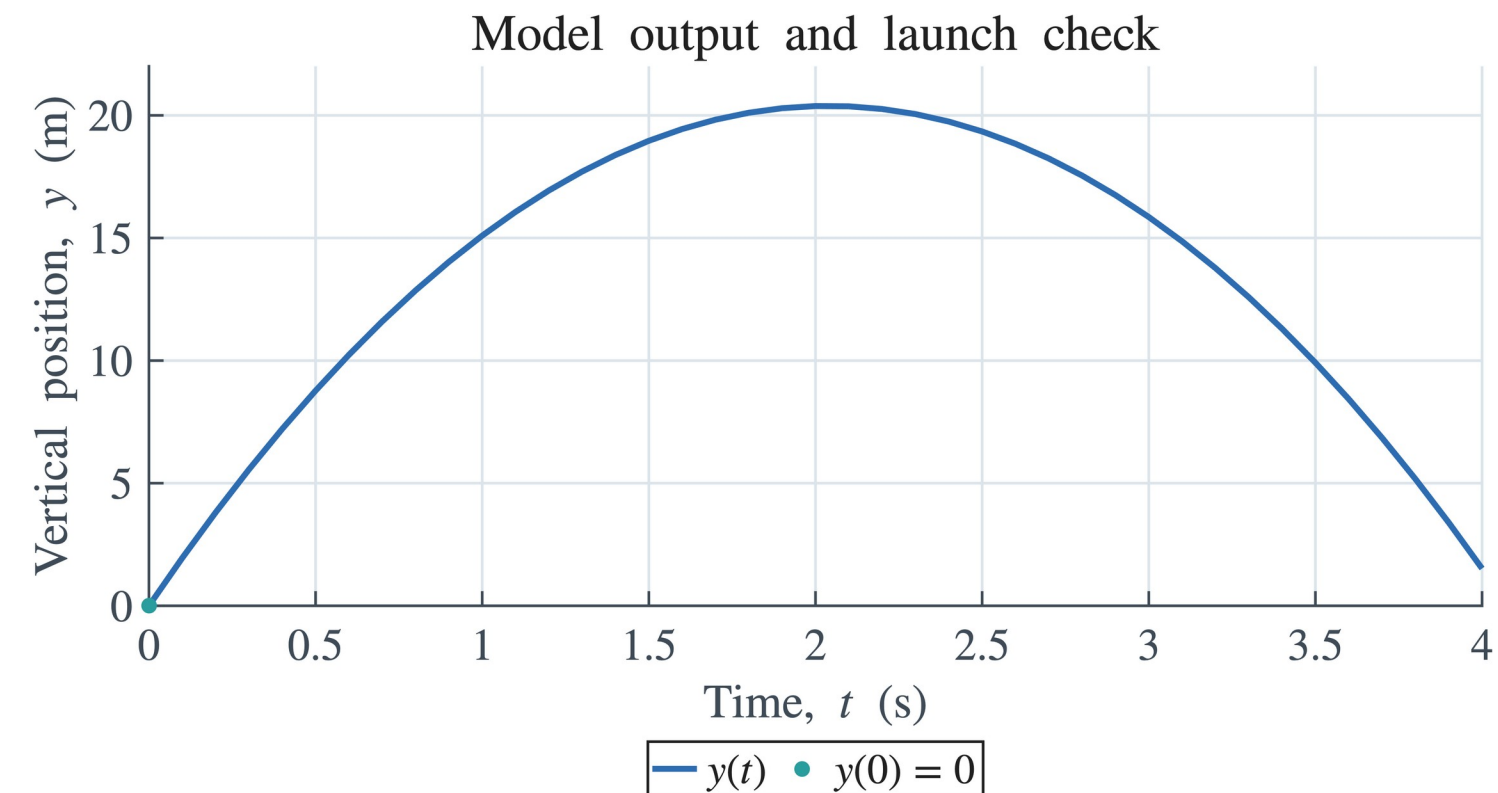
Checkpoint: explain why the model line contains `.*` and `.^`

Run the model, then read the graph

Prediction comes before the plot; a known value checks the result

Trace this short fragment

```
t_s = 0:0.1:4;  
y_m = 20.*t_s - 0.5*9.81*t_s.^2;  
plot(t_s,y_m,'LineWidth',2)  
xlabel('Time (s)')  
ylabel('Vertical position (m)')  
assert(abs(y_m(1)) < 1e-12)
```



Prediction: rise, one peak, then fall. Check: $y_m(1) = 0$ m

Week 1 exit ticket

Use the same ball model and answer in one or two sentences

- 1 Name the output variable and its unit

- 2 What does `t_s(11)` select?

- 3 Which operator lets MATLAB square every value in `t_s`?

- 4 State one prediction for the graph and one value you can validate

Next time: turn the same physical model into a traceable algorithm

Working exposure: selected values

Read a few model outputs before drawing the full graph

For the same vertical-motion model, each selected time has one computed height.

0.0 s 0.00 m launch	1.0 s 15.10 m rising	2.0 s 20.38 m near the peak	3.0 s 15.86 m falling	4.0 s 1.52 m near return
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Check the pattern: height rises, reaches one maximum, then falls toward the launch height.